

JASC DATA PROCESSOR

Comprehensive User Manual

A lightweight, modular desktop application for data ingestion, visualization, mathematical transforms, statistics, and curve fitting.

Reference Manual & Guide

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Table of Contents

1. Installation & Getting Started **3**

System Requirements 3

Launching the Application 3

2. User Interface Layout **4**

3. Data Ingestion **5**

CSV/TXT File Import 5

Project Files 5

Manual Spreadsheet Entry 5

Digitizing Data from Images (Import from Figure) 5

4. Data Management & Sheet Editing **7**

Spreadsheet Cell Operations 7

Inserting & Deleting Rows and Columns 7

Column Metadata (Roles, Units, Long Names) 7

Data Preparation Tools 7

5. Mathematical Formulas & Expressions **9**

Using the Apply Formula Dialog 9

Positional Aliases and Namespaces 9

6. Calculus, Signal Processing & Interpolation **10**

Numerical Differentiation & Integration 10

Smoothing Workbench 10

FFT and Inverse FFT 10

Interpolation Workbench 10

7. Statistics Workbench **12**

Single-Column Summaries & Diagnostics 12

Two-Column & Paired Tests 12

Grouped & Factor Analysis 12

Multi-Column PCA & Correlations 12

Companion Plots 12

8. Curve Fitting Engine **14**

Standard Curve Fitting 14

Custom Mathematical Fit Models 14

Uncertainties & ODR (Orthogonal Distance Regression) 15

9. Visualization & Saved Plots **16**

Plot Series Configuration 16
Axis Scaling, Ranges & Custom Styling 16
Legend and Live Preview 16
Saved Plot Definitions 17
10. Batch Processing Workbench 18
Running Batch Operations 18
Execution Controls & Overwrite Safeguards 18
11. Project Files, Export, and Persistence 20
Saving and Loading Projects 20
Exporting Data and Figures 20
Large Datasets 20
12. Troubleshooting & FAQs 21

1. Installation & Getting Started

System Requirements

- Python 3.10 or higher
- Dependencies listed in requirements.txt (including PySide6, pandas, numpy, scipy, matplotlib, odrpack, and SciencePlots).

Launching the Application

Navigate to the directory containing JASC in your terminal, and run:

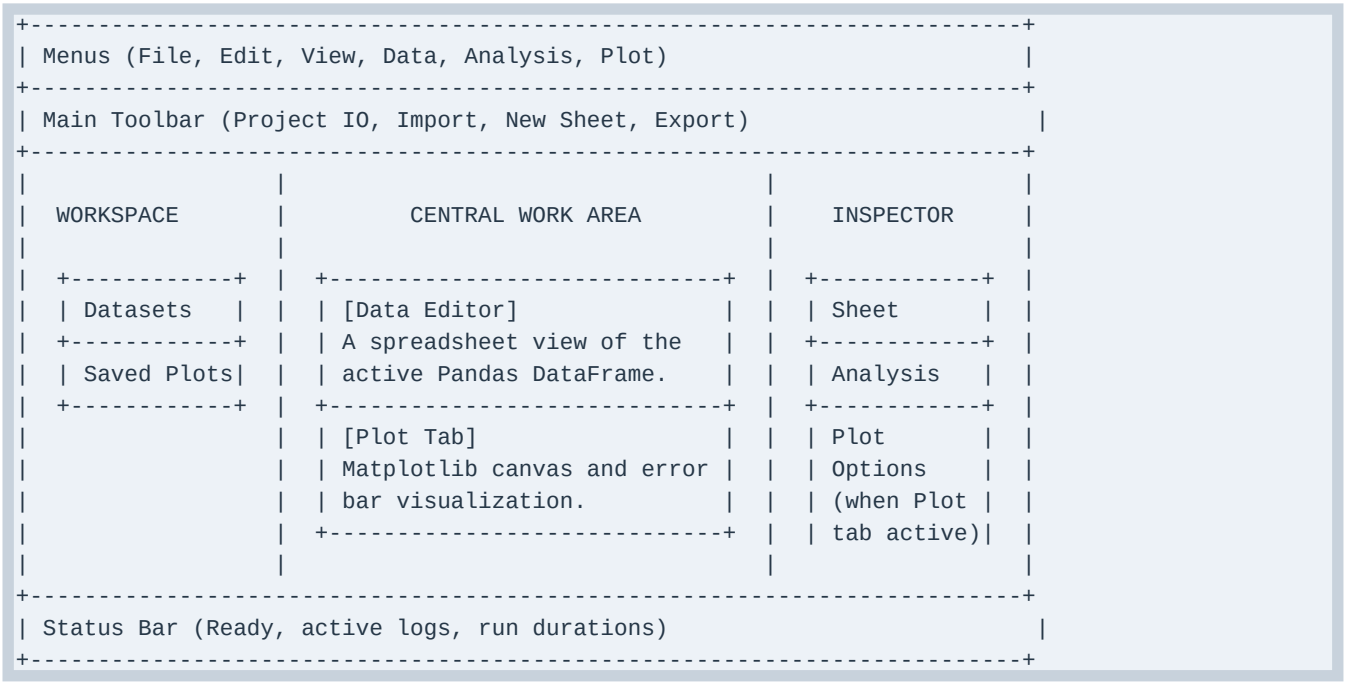
```
python main.py
```

This initializes the application window and opens an empty workspace.

---

2. User Interface Layout

JASC uses a Docked Workspace Layout where all secondary control panels dock neatly around the main work area.



- Central Work Area: Contains two tabs:

1. Data: Displays the selected sheet in a spreadsheet view.
2. Plot: Displays the active visualization canvas.

- Workspace Panel (Left): Displays list of active Datasets and Saved Plots. Double-clicking items swaps active view.
- Plot Inspector (Right): Appears in the Plot tab and contains series styling, axes settings, and locator coordinates. It is hidden in the Data tab because worksheet, preparation, and analysis operations are available from the upper menus and table context menus.
- Resetting Layout: If panels get closed or misplaced, select View > Reset View to Default (Ctrl+Shift+R) to restore the original layout.

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3. Data Ingestion

CSV/TXT File Import

1. Select File > Import Data... (Ctrl+I) or click the Import button in the main toolbar.
2. In the dialog, customize:
- Delimiter: Comma, Tab, Semicolon, or Custom whitespace.
- Headers: Detect headers automatically or specify start row.
3. The dataset appears in the Workspace panel on the left with a unique name.

Project Files

Use File > Save Project... to store the complete workspace as a .jasc project file. Project files preserve imported datasets, manually-created sheets, formulas, column metadata, custom fit models, and saved plot definitions. Use File > Open Project... to restore that workspace later.

Manual Spreadsheet Entry

1. Select File > New Blank Sheet (Ctrl+N), click the toolbar icon, or click New Sheet at the bottom of the Datasets list.
2. Add or insert rows and columns from the Data menu or the table's context menus.
3. To generate data patterns, use Data > Fill Range. Specify:
- Target Column
- Start Value and Step Size
- Row Range and Row Count
- Click OK to populate arithmetic series (e.g. timestamps or independent variables).

Digitizing Data from Images (Import from Figure)

If you have a plot in an image (JPG/PNG) but need the underlying numbers, use JASC's built-in Figure Digitizer.

1. Launch it via File > Import From Figure... or the main toolbar button.
2. Load an image using Load Image on the top toolbar.

If the figure was photographed at an angle or scanned with perspective skew, correct the image before calibration:

1. Open the Correction tab.
2. Pick the four corners of the plotted frame or the axis-bar rectangle: top-left, top-right, bottom-right, and bottom-left.

3. Review the green quadrilateral on the canvas and adjust the corrected image width/height if needed.
4. Click Apply Correction. The canvas is replaced with a rectified working image.

After correction, all calibration, manual picking, automatic tracing, and symbol detection run against the corrected image. Use Reset Original to return to the uncorrected image.

Before digitization, calibrate the pixel-to-data mapping:

1. Go to the Calibration tab.
2. Click Pick X1 and click on a known value on the image's X-axis. Enter its calibrated value.
3. Repeat for Pick X2, Pick Y1, and Pick Y2.
4. Select whether the axis is Linear or Logarithmic.
5. Use the Precision Zoom panel with the magnifier enabled to click exact pixel centers.

- Select the Manual mode on the mode banner.
- Click anywhere on the plot lines to manually place points.
- To add error bars / uncertainties, click Set Uncertainty for a selected point, and click either a radial boundary or four coordinates representing error limits. Manual uncertainties are rendered as blue error bars.

If the plot has solid lines, let JASC trace it:

1. Set a painted region using the ROI Brush (translucent green mark) to restrict tracing.
2. Select Curves/Lines > Auto Trace.
3. Choose Dark Pixels or Color Sample (click a color in the image using the sampler).
4. Click Run Auto Trace to extract pixel paths.

For scatter plots:

1. Go to Symbols.
2. Click Sample Symbol and drag a bounding box around a symbol in the image.
3. JASC learns the template symbol, size, and color.
4. Select a Symbol Shape Prior (e.g., Circle, X cross, Plus cross, Square) to filter lines touching the symbols.
5. Click Run Symbols Match.
6. The digitizer extracts symbol positions along with pixel and calibrated uncertainty values based on symbol size.

Click Create Dataset to save your digitized coordinate table into the JASC workspace.

---

4. Data Management & Sheet Editing

Spreadsheet Cell Operations

- Cell Editing: Double-click any cell in the Data Editor to change its value. Press Enter to commit.
- Undo/Redo: Supported for all sheet changes (cell edits, deletes, formula applications, etc.). Press Ctrl+Z to Undo, and Ctrl+Y to Redo.
- Clipboard Copy/Paste:
- Select cells and press Ctrl+C to copy.
- Click on a destination cell and press Ctrl+V to paste.
- JASC automatically expands rows or columns to accommodate pasted blocks.

Inserting & Deleting Rows and Columns

Right-click on any cell or header in the Data Editor to access the spreadsheet context menu:

- Insert Row / Column: Adds a row/column before the selection.
- Delete Row / Column: Deletes selected rows or columns.
- Whole-Column Clipboard: Copy and paste entire columns, which preserves metadata (like column roles and units).

Column Metadata (Roles, Units, Long Names)

Double-click a column header (e.g. col_0) to open the Column Metadata dialog.

- Name: The column identifier shown in the table and used in formulas.
- Role: Select x, y, z, x_error, y_error, group, label, or weight.
- Type: Leave as Auto, or mark the column as Numeric, Scientific, or Text. Numeric and Scientific columns reject non-numeric edits and are validated before math operations.
- Display Precision: Control fixed-point or scientific display precision without changing the stored value.
- Long Name & Unit: Used to automatically prefill Matplotlib plot titles and axis labels.
- Comments: Document column origin or transformations. JASC also writes operation provenance here for generated columns.

Data Preparation Tools

The Data menu includes reusable tools for shaping datasets before analysis:

- Filter / Query: Filter rows by equality, numeric comparisons, ranges, contains/starts-with text checks, comma-separated list membership, blank, or non-blank values.
- Sort: Sort by one or more columns in ascending or descending order.
- Remove Duplicates: Choose the columns that define a match, then keep the first occurrence, keep the last occurrence, or remove every row belonging to a duplicate group.
-

Missing Values: Drop rows with missing selected values, fill with a constant, fill with column mean/median, or interpolate numeric gaps.

- Bin / Average: Split a numeric bin column into equal-width bins and calculate count, mean, standard deviation, SEM, minimum, and maximum for selected value columns.
- Group / Aggregate: Group by one or more factor columns and compute count, mean, standard deviation, SEM, minimum, maximum, and sum for selected numeric columns.
- Merge / Join: Join two datasets with inner, left, right, or outer join behavior using selected key columns.
- Reshape: Transpose a dataset or melt selected value columns into a long/tidy table.

Each preparation tool creates a new result sheet so the source data remains available.

---

5. Mathematical Formulas & Expressions

Apply custom expressions directly in columns.

Using the Apply Formula Dialog

1. Select Data > Apply Formula... (Ctrl+Shift+F).
2. Select the target column (or enter a new name to create one).
3. In the formula text box, type your expression (e.g., col("y") * 2.5 or col1 / col2).
4. Select the cell scope: All Rows or Selected Rows Only.
5. Click Run to execute.

Positional Aliases and Namespaces

- Positional Aliases: You can refer to columns by their index, like col1 (first column), col2 (second column), etc., as an alternative to writing the exact name col("Column Name").
- NumPy/SciPy Namespace: JASC sandbox imports standard mathematical packages. You can write expressions using:
- np.sin(), np.log10(), np.exp(), np.sqrt(), np.pi
- special (SciPy special functions)
- stats (SciPy stats distributions)
- np.random.normal() (NumPy random values)
- Row Helpers: row is a one-based row-number series, and index exposes the DataFrame index as a series.
- Exact Name Helper: Use col("Column Name") for headers containing spaces, punctuation, or Python keywords.

> **[WARNING]** Standard Python power syntax is `**` (e.g. `col1 ** 2`). Do not use `^` as JASC blocks it to prevent common XOR bitwise operation bugs.

---

6. Calculus, Signal Processing & Interpolation

Numerical Differentiation & Integration

Available via the Analysis menu:

- Derivative: Select X and Y columns. Choose from methods: *NumPy gradient*, *Forward difference*, *Backward difference*, *Central difference*, or *Savitzky-Golay* smoothing derivative.
- Integral: Select X and Y columns. Choose from integration algorithms: *Trapezoid*, *Simpson*, *Left*, *Right*, or *Midpoint* rectangle. You can specify an Initial Value offset (integration constant) to align the integrated curve with physical bounds.

Select Plot result immediately to automatically view the calculus output.

Smoothing Workbench

Open Analysis > Smoothing... to preview and write a smoothed signal into a worksheet column. Available methods are adjacent average, moving median, Savitzky-Golay, LOWESS/LOESS, FFT low-pass smoothing, and derivative smoothing followed by integration. The dialog supports multiple Y columns, multiple compatible worksheets, name/position column matching, safe output-name conflict policies, cancelable batch progress, and an immediate original-versus-smoothed comparison plot.

Derivative smoothing + integration is useful when the measured curve is noisy but its underlying rate of change is expected to vary smoothly. JASC calculates dy/dx, optionally applies a moving-median spike prefilter, then lets you smooth the derivative with any included smoothing method: adjacent average, moving median, Savitzky-Golay, LOWESS/LOESS, or FFT low-pass. It integrates the result from the original first value. Preserve first and last data values distributes residual integration drift across the reconstructed curve so both endpoints remain anchored. Savitzky-Golay and FFT require evenly spaced X values, including when selected as the derivative smoother.

FFT and Inverse FFT

- FFT (Fast Fourier Transform): Computes the spectral amplitude and frequency. Select a signal column, sampling interval, and a window function (*Rectangular*, *Hann*, *Hamming*, *Blackman*, *Bartlett*, *Flat top*). JASC creates a new sheet containing the frequency spectrum.
- iFFT (Inverse FFT): Reconstructs time-domain signals. Select the real and optional imaginary spectrum columns.

Interpolation Workbench

Launch via Analysis > Interpolation... to predict values at arbitrary points:

- 1D Interpolation: Linear, nearest neighbor, cubic spline, PCHIP, Akima, or smoothing splines.

- 2D Interpolation: Nearest, linear scattered, bivariate cubic spline, Clough-Tocher piecewise cubic, smooth bivariate spline, and RBF kernels.
- N-D Interpolation: Nearest, linear scattered, and RBF kernels for coordinate sets with two or more dimensions.
- RBF Kernels: Linear, thin-plate spline, cubic, and Gaussian. Gaussian RBF can use an automatic or user-entered epsilon value.
- Query Coordinates: Generate a query sequence automatically, select existing query columns, or enter coordinates manually. Generated 2D queries create a full X/Y grid.
- Outside Range Policy: Return blanks outside the source range, extrapolate when the chosen method supports it, or clamp 1D predictions to the nearest source edge value.
- Multiple Value Columns: Select more than one value column to create prediction columns for each signal in the result sheet.

---

7. Statistics Workbench

Perform statistical evaluations directly inside the JASC environment. Access these under Analysis > Statistics...

Single-Column Summaries & Diagnostics

Returns descriptive parameters (count, missing count, mean, median, standard deviation, variance, SEM, min/max, range, coefficient of variation), percentiles, distribution shape (skewness and kurtosis), normality tests (Shapiro-Wilk, D'Agostino-Pearson, Jarque-Bera, Anderson-Darling), confidence intervals for the mean, Z-score series, cumulative statistics, and outlier checks using Z-score and IQR fences.

Two-Column & Paired Tests

Evaluate relationships between columns:

- Correlations: Pearson, Spearman, and Kendall coefficients.
- Comparison Tests: Paired t-tests, Wilcoxon signed-rank, Welch independent-samples t-test, Mann-Whitney U, Kolmogorov-Smirnov, Levene, and Bartlett tests.
- Agreement and Error: Bland-Altman statistics, covariance, linear regression, MAE, RMSE, mean error, and R^2 against the primary column.

Grouped & Factor Analysis

If you have a factor or group column (e.g. categorical variables) and a value column:

- Summarizes statistics per group.
- Runs One-way ANOVA or Kruskal-Wallis tests.
- Performs Tukey HSD post-hoc comparisons and pairwise Mann-Whitney tests with Holm-Bonferroni adjusted p-values.

Multi-Column PCA & Correlations

Analyze multiple variables in a single operation:

- Computes pairwise correlation matrices.
- Performs Principal Component Analysis (PCA), reporting eigenvalue distribution, explained variance, loading vectors, and PC scores.

Companion Plots

Check Create companion plot to generate Matplotlib visualizations linked to your stats:

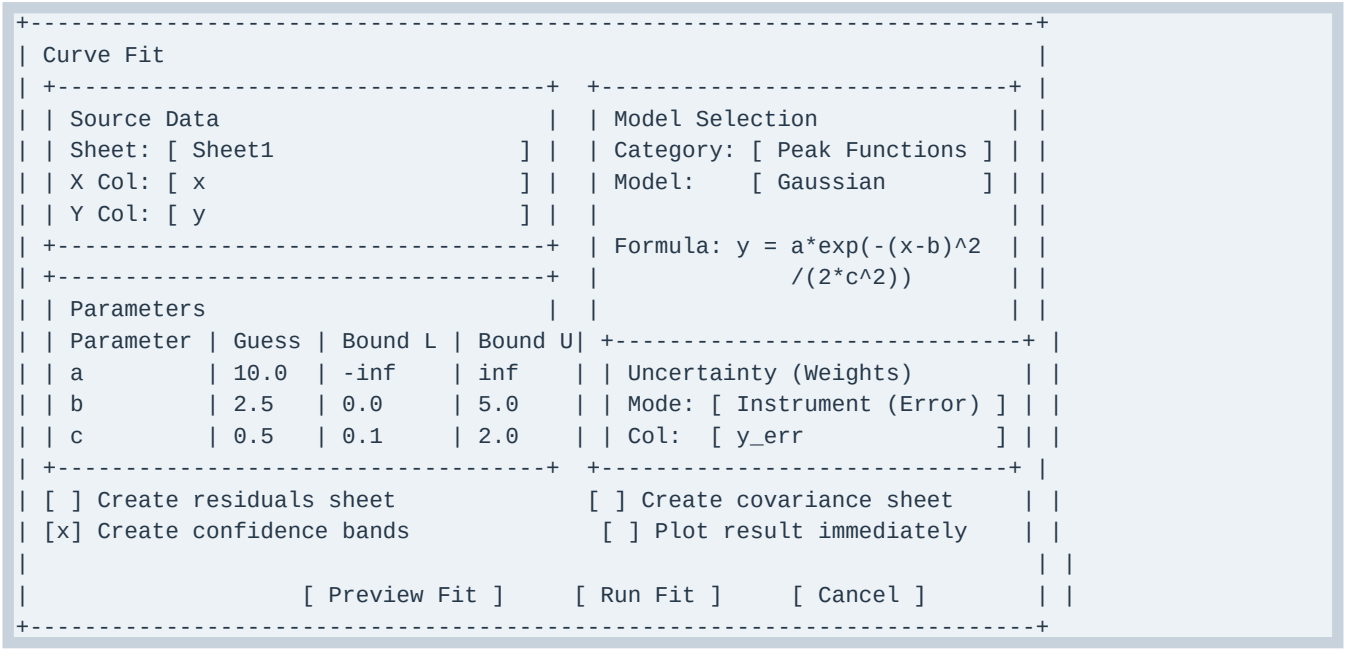
- Normality histograms with normal distribution fits.
- Grouped box/violin comparison charts.
- Bland-Altman bias/limit charts.

- PCA scree plots.

---

8. Curve Fitting Engine

JASC provides a curve fitting engine for analytical models. Open via Analysis > Curve Fit... (Ctrl+Shift+C).



Standard Curve Fitting

1. Select the source Sheet, X, and Y columns.
2. Select a model such as Linear, Quadratic, Cubic, Exponential growth/decay, Double exponential decay, Gaussian, Lorentzian, Pseudo-Voigt, Sine, Damped sine, Logistic, Gompertz, Error-function step, Power law, Log-normal, Weibull, Heidler, Bessel J/Y, Stretched exponential, or Asymmetric Gaussian.
3. Set initial parameter guesses, lower and upper bounds, or check Fixed to hold a parameter constant.
4. Select fit algorithms: *Levenberg-Marquardt*, *Trust Region Reflective*, *Dogbox*, or *Orthogonal Distance Regression*.
5. Click Run Fit. JASC displays a Fit Report with parameters, fixed/free state, bounds, standard errors, covariance, R^2, adjusted R^2, RMSE, MAE, residual mean/std, warnings, and optional confidence bands.

Custom Mathematical Fit Models

Create custom fitting equations via Analysis > Custom Fit Models...:

1. Click Add Model.
2. Define parameters as a comma-separated list (e.g. a, b, c).

3. Write the math expression using safe Python/NumPy (e.g. a * np.sin(b * x + c)).
4. Save the model. It becomes available in the standard Curve Fit model selector under the *Custom* category.

Uncertainties & ODR (Orthogonal Distance Regression)

- Y Uncertainty: Select a column representing Y measurement error to run weighted fitting.
- X/Y Uncertainty (ODR): If your independent variable X also has errors, select an X-error column. JASC automatically swaps the solver to Orthogonal Distance Regression (ODR) to calculate correct parameters accounting for errors in both dimensions.
- Weight Modes: Interpret uncertainty columns as sigma/error values or as weights. Absolute sigma controls whether covariance is scaled by residual variance.
- Result Sheets: Fits can create fitted-curve, residual, covariance, parameter-summary, and batch-summary datasets depending on the workflow.

---

9. Visualization & Saved Plots

Plot Series Configuration

1. Select the Plot tab, then view the Plot Inspector on the right.
2. Configure a series and click Add as New in the Series tab. Selecting an existing queued series live-binds the editor to it; changing its subtype, data mapping, or appearance updates that queued series and participates in Auto preview. Apply to Selected remains available for an immediate explicit update.
3. Assign columns to X, Y (and Z/weights if applicable).
4. Select a Plot Type:

- *2D Line & Scatter*: Line, Scatter, Line + Markers, Errorbar.
- *Distribution & Reporting*: Box Plot, Violin Plot, Histogram, Bar (Vertical/Horizontal), Area/Fill.
- *Composition*: Pie, Donut, Voronoi Composition.
- *2D Surface Maps*: Interpolated Heatmap, Contour, Filled Contour.

5. Set styles: line color, line width, markers, fills, and alphas.

Axis Scaling, Ranges & Custom Styling

- Go to the Axes tab in the Plot Inspector:
- Manually set limits for X, left Y, and right Y axes, or leave them blank to auto-scale.
- Swap scales between Linear and Logarithmic.
- Toggle the plot grid on/off.
- In the Axes tab's style controls:
- Select SciencePlots journal templates (e.g. science, nature, ieee).
- Enable/disable LaTeX formatting.
- Adjust output figure dimensions.
- Use locators/cursors to find Y from a supplied X or every X intersection for a supplied Y. Surface plots accept X and Y coordinates and return interpolated Z. Horizontal curve segments at the requested Y are reported as ambiguous rather than producing an arbitrary X.

Legend and Live Preview

Use the dedicated Legend tab to control legend visibility, inside or outside placement, text and title sizes, optional title, column count, frame visibility and opacity, marker scale, handle length, and whether locator markers or confidence bands appear. A series can be excluded individually with Show this series in legend. Entries from the left and right Y axes follow queued-series order and duplicate labels are removed.

Global axes, style, figure-size, and legend changes update through a short debounced Auto preview by default. Turn Auto preview off for expensive plots and click Apply when ready. Invalid range text marks the preview as out of date without interrupting typing with modal dialogs.

Heatmap, Contour, and Filled Contour require at least three distinct, non-collinear X/Y coordinate points and numeric Z values. If the selected coordinates cannot form a 2D interpolation surface, the canvas displays the rendering reason rather than an empty white plot.

Saved Plot Definitions

New / Empty Plot is the default plot workspace entry. It starts a clean unsaved plot with default axes and styling; analysis-generated plots also become unsaved drafts so they cannot accidentally overwrite a previously loaded saved plot.

When New Empty Plot would replace a draft with unsaved changes, JASC asks whether to Save, Discard, or Cancel. Cancel keeps the current plot intact.

Once your plot looks correct, save its state so you can return to it later:

1. Click Save Plot at the top of the Plot tab or select Plot > Save Current Plot...
2. Name the plot definition.
3. The plot appears in the Plots tab in the Workspace panel.
4. Double-clicking any item in this list restores the exact series, styling, axes, scales, and grids.

---

10. Batch Processing Workbench

JASC's most powerful productivity feature is the Batch Processing Workbench, which lets you repeat calculations across multiple worksheets.

```
+-------------------------------------------------------------------------+
| Batch Preflight Preview                                                 |
|                                                                         |
| Target Sheet | Action   | Resolved Inputs  | Outputs       | Status      |
|--------------+----------+------------------+---------------+-------------|
| Dataset_A    | Run      | col1, col2       | col_res       | OK          |
| Dataset_B    | Run      | col1, col2       | col_res       | OK          |
| Dataset_C    | Skip     | col1 (missing)   | --            | Err: col2   |
|                                                                         |
|                     [ Run Batch ]        [ Cancel ]                     |
+-------------------------------------------------------------------------+
```

Running Batch Operations

Every main operation dialog (Formula, Derivative, Integral, FFT, Interpolation, Statistics, Curve Fit) supports batch execution:

1. In the operation dialog, expand the Target Worksheets list at the bottom.
2. Select the sheets you want to target.
3. Select the Column Binding Mode:

- Exact Name: Matches columns by exact text headers.
- Column Position: Matches columns by their layout order (e.g. col1, col2).

4. Click Run. JASC displays a Batch Preflight Preview listing target sheets, mapping resolutions, outputs, and any compatibility errors.
5. Confirm and run. A cancelable modal progress bar shows the execution state.

Execution Controls & Overwrite Safeguards

- Overwrite Safeguards: In batch formulas or calculus, JASC defaults to creating new unique output columns to prevent accidental data loss. You can explicitly confirm an overwrite update policy in the preflight check.
- Batch Summaries: Multi-sheet calculations generate a cumulative summary sheet (e.g., {active_sheet}_fit_batch_summary or {active_sheet}_statistics_batch_summary). These summary sheets aggregate parameter fits, R^2, averages, test metrics, and execution logs in a tidy table format ready for plotting or exporting.
- Parallel Workers: JASC utilizes process-based multiprocessing for headless batch sheets, running operations in parallel across CPU cores without locking the active GUI.
-

Naming Tokens: Use {source} and, where supported, {column} in output names to create predictable per-sheet or per-column result names.

- Column Binding: Exact-name binding is safest when every sheet shares headers. Position binding is useful for repeated instrument exports where the first, second, and third columns have consistent meaning but different names.

---

11. Project Files, Export, and Persistence

Saving and Loading Projects

Project files use the .jasc extension. They are ZIP archives containing CSV data and JSON metadata. Saving a project preserves:

- Datasets and source paths.
- Column names, roles, units, long names, comments, value types, and display precision.
- Column formulas and operation provenance comments.
- Saved plot definitions.
- Project custom fit models.

Exporting Data and Figures

Use File > Export Dataset... to write the active dataset as CSV/TXT. Plot figures are exported from the Matplotlib toolbar in the Plot tab. Use the saved-plot system when you want to preserve plot configuration inside the project rather than only exporting a static image.

Large Datasets

JASC profiles dataset size and memory use. Column widths are calculated from a bounded row sample rather than scanning a complete worksheet. Clipboard data and generated ranges are parsed, grown, and assigned as column blocks, so large pastes do not append or update the DataFrame one cell at a time.

Undo history retains at most 20 worksheet snapshots within a 256 MiB budget per undo/redo stack. If the current worksheet is too large for one snapshot, JASC avoids making the expensive copy and shows a status-bar warning that the edit cannot be undone. Very large file and project parsing is still an in-memory Pandas operation; background/cancelable import and out-of-core datasets remain planned.

---

12. Troubleshooting & FAQs

Make sure you are executing python from the active conda environment:

```
& "C:\Users\julia\anaconda3\envs\dataProcessing\python.exe" main.py
```

JASC warns before importing a delimited file of 100 MiB or more and flags a loaded worksheet at 2,000,000 cells or an estimated 250 MiB of DataFrame memory. The Data Editor samples at most 25 rows when sizing columns above the 250,000-cell responsiveness threshold.

- Check if your initial guesses are realistic.
- Verify if bounds are too narrow or overlap.
- Swap from *Levenberg-Marquardt* to *Trust Region Reflective* to respect parameter bounds.

To save a sheet as a CSV or TSV file, select the sheet and choose File > Export Dataset.... To export figures, click the save floppy disk icon on the Matplotlib toolbar in the Plot tab.